

Pt100

Installation and operating instructions

(GB) (D) (F) (E) (DK) (PL) (RU) (EE)



Pt100

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Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. General

1.1 Applications

The Pt100 is a temperature-dependent resistance sensor, suitable for fitting in a submersible motor.

A Pt100 sensor allows

- continuous monitoring of the motor temperature and
- protection against too high motor temperature.

The Pt100 sensor is used for temperature monitoring of

- Grundfos MS 6000 motors
- Grundfos MMS 6000, 8000, 10000 and 12000 motors.

Note: When fitted in an MS 6000 motor with defective temperature transmitter (Tempcon) and connected to a control unit, the Pt100 makes it possible to re-establish monitoring and protection of the motor.

1.2 Frequency converter

The Pt100 sensor allows monitoring of the temperature in MS 6000 motors connected to a frequency converter.

2. Technical data

The Pt100 sensor is supplied with an PR 5714 relay.

The technical specifications for the PR 5714 relay can be found in the installation and operating instructions supplied with the relay.

3. Installation

3.1 MS 6000 motors with Pt100 sensor

The Pt100 sensor is placed in the special staybolt supplied with the Pt100 kit. The sensor will thus acquire approximately the same temperature as the motor liquid during operation.

3.1.1 MS 6000, model A

To fit the Pt100 sensor, proceed as follows:

See fig. A on page 39.

1. Separate the motor from the pump.
2. Check the motor liquid level.
3. Remove one of the staybolts close to the plug, pos. A. Reuse the nut, pos. B.
4. Fit the special staybolt, pos. C, with the through-going hole pointing outwards and tighten it to 50 Nm.
5. Fit the motor to the pump. Tighten the bolts diagonally to 50 Nm.
6. Place the sensor with the shrink-on sleeve, pos. D, in the hollow staybolt, pos. C.
7. Heat the sleeve so that there is a connection between the staybolt thread and the sensor cable.
8. **Note:** Instead of the shrink-on sleeve, it is possible to use resin (part No. ID1302) with a hardening time of 12 hours.
9. Check whether the sensor cable and the motor cable can be placed in the cable guard. If not, fit the sensor cable on the side of the cable guard so that the cable is protected against damage when the pump is lowered into the borehole.
10. Fit the cable guard.
11. Fasten the sensor cable to the motor cable with cable clips.

3.1.2 MS 6000, model B

To fit the Pt100 sensor, proceed as follows:

See fig. B on page 39.

1. Slacken the four hexagon-head screws in the motor top.
2. Check the motor liquid level.
3. Remove one of the hexagon-head screws close to the plug, pos. A.
4. Fit the special staybolt in the motor top, pos. C.
5. Fit the motor to the pump. Tighten the screws diagonally to 50 Nm.
6. Pull the union nut, pos. B, and the rubber gasket, pos. D, up behind the sensor and then press the sensor to the bottom of the bolt, pos. C.
7. Tighten the union nut so that the sensor cable is secured by the rubber gasket.
8. Check whether the sensor cable and the motor cable can be placed in the cable guard. If not, fit the sensor cable on the side of the cable guard so that the cable is protected against damage when the pump is lowered into the borehole.

- 9. Fit the cable guard.
- 10. Fasten the sensor cable to the motor cable with cable clips.

3.2 MMS motors with Pt100 sensor

From factory, the motors are provided with a blanking plug, see fig. 1, 2 or 3.

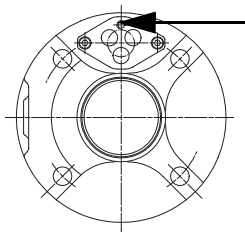


Fig. 1 Position of blanking plug – MMS 6000 and MMS 8000

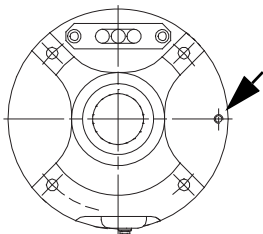


Fig. 2 Position of blanking plug – MMS 10000

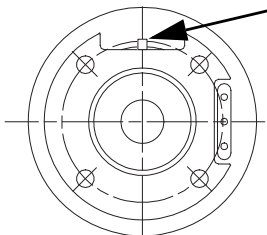


Fig. 3 Position of blanking plug – MMS 12000

3.2.1 Adjustment of the sensor sleeve for MMS motors

The length of the stainless-steel sleeve used for fitting the Pt100 sensor in MMS motors must be adjusted according to the lengths stated in the table.

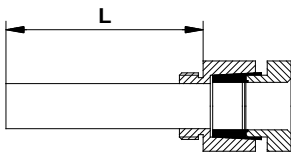


Fig. 4 Adjustment of sensor sleeve

Motor	L (sleeve)	Insertion depth
MMS 6000	–	100 mm
MMS 8000	–	
MMS 10000	100 mm	
MMS 12000	35 mm	

3.2.2 MMS 6000 and MMS 8000

To fit the Pt100 sensor, proceed as follows:
See fig. C on page 40.

1. Separate the motor from the pump.
2. Check the motor liquid level.
3. Slacken the cable entry and remove the screw, pos. A.
4. Fit the Pt100 sensor using the 250 mm shrink-on sleeve.
5. Heat the sleeve in the cable entry.
6. Tighten the cable entry. Check that the Pt100 sensor is fixed.
7. Tilt the motor until the screw in the filling hole is the highest point of the motor.
8. Inject motor liquid into the motor.
9. Fit the motor to the pump.
10. Check whether the sensor cable and the motor cable can be placed in the cable guard.
If not, fit the sensor cable on the side of the cable guard so that the cable is protected against damage when the pump is lowered into the borehole.
11. Fit the cable guard.
12. Fasten the sensor cable to the motor cable with cable clips.

3.2.3 MMS 10000 and MMS 12000

To fit the Pt100 sensor, proceed as follows:
See fig. D or E on page 41.

1. Separate the motor from the pump.
2. Remove the blanking plug, pos. A.
The blanking plug of MMS 10000 motors is positioned vertically.
The blanking plug of MMS 12000 motors is positioned horizontally.
3. Check the motor liquid level.
4. Fit the sensor in the sleeve, see section 3.2.1 *Adjustment of the sensor sleeve for MMS motors* and fig. 5.

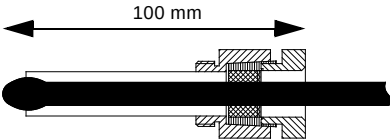


Fig. 5 Pt100 sensor fitted

5. Fit the sensor in the motor, pos. B.
6. Tilt the motor until the screw in the filling hole is the highest point of the motor.

7. Inject motor liquid into the motor.
8. Fit the motor to the pump.
9. Check whether the sensor cable and the motor cable can be placed in the cable guard. If not, fit the sensor cable on the side of the cable guard so that the cable is protected against damage when the pump is lowered into the borehole.
10. Fit the cable guard.
11. Fasten the sensor cable to the motor cable with cable clips.

4. Electrical connection

The electrical connection must be carried out by an authorised electrician in accordance with local regulations.



Before making any connections, make sure that the electricity supply has been switched off.

Note: Make sure that the PR 5714 relay is suitable for the electricity supply on which it will be used.

4.1 Connection of PR 5714 relay



Connect the PR 5714 relay as described in the installation and operating instructions supplied with the relay.

Connect the PR 5714 relay to the Pt100 sensor as illustrated in the wiring diagram in fig. 6.

Note: The PR 5714 can be connected to an external alarm device for alarm indication in case the stop or warning limits are exceeded.

4.1.1 Operating functions of the PR 5714 relay

During operation, the motor temperature is indicated on the display.

Check the function of the Pt100 sensor as follows:

1. Read the temperature on the display.
2. Stop the motor.
3. Check that the motor temperature drops.

If the motor temperature does not drop during the interruption of operation, the cause may be one or more of the following faults:

- Sensor cable breakage
- Defective Pt100 sensor
- Defective PR 5714 relay.

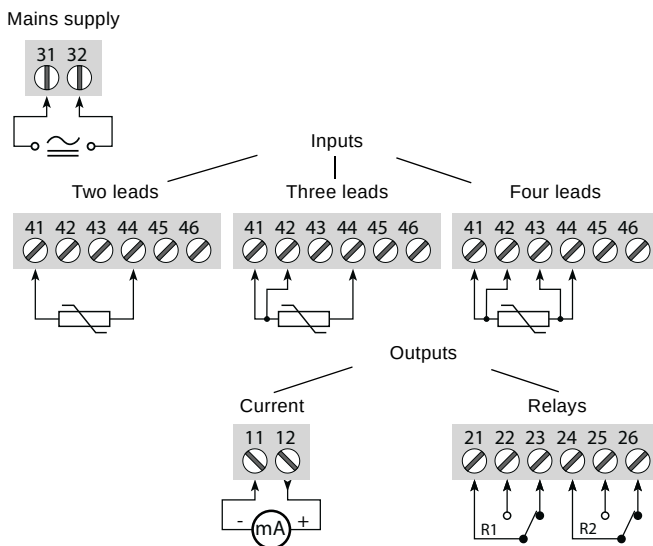


Fig. 6 Connection of PR 5714 relay

5. Fault finding chart



Before any dismantling, make sure that the electricity supply has been switched off and that it cannot be accidentally switched on.

GB

Fault	Cause
1. The temperature indication on the PR 5714 relay does not drop even if the motor has been stopped for more than 5 minutes.	a) The Pt100 sensor is defective. Check the sensor cable. The resistance values below apply to a standard sensor cable: White lead to white lead: 0-10 Ω – OK. Red lead to red lead: 0-10 Ω – OK. White lead to red lead: 100-150 Ω – OK. b) The PR 5714 relay is defective.
2. The PR 5714 relay does not indicate a temperature within the range of 0-100°C.	a) The Pt100 sensor is defective. b) The PR 5714 relay is defective.

6. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

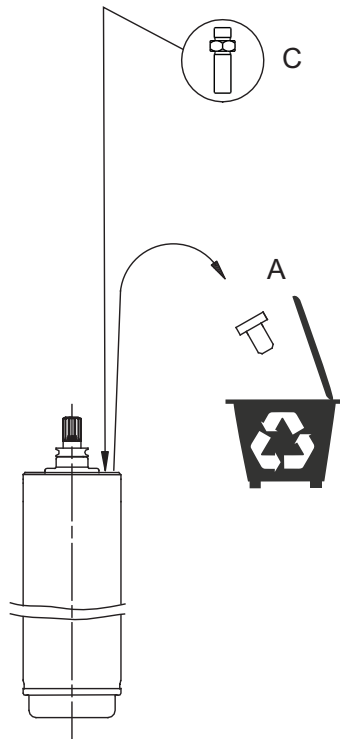
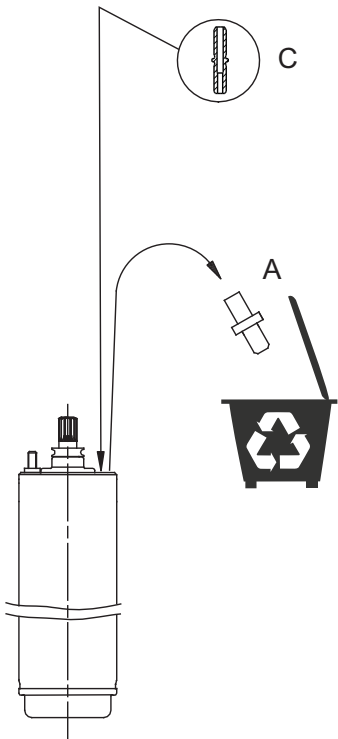
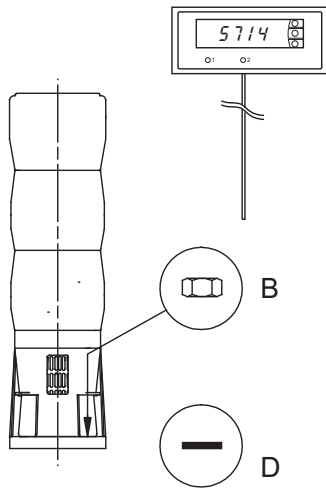
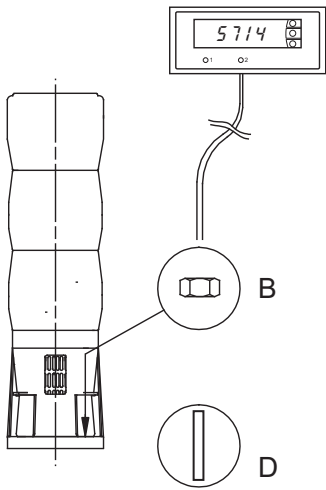


Fig. A MS 6000, model A

Fig. B MS 6000, model B

TM03 5452 3606

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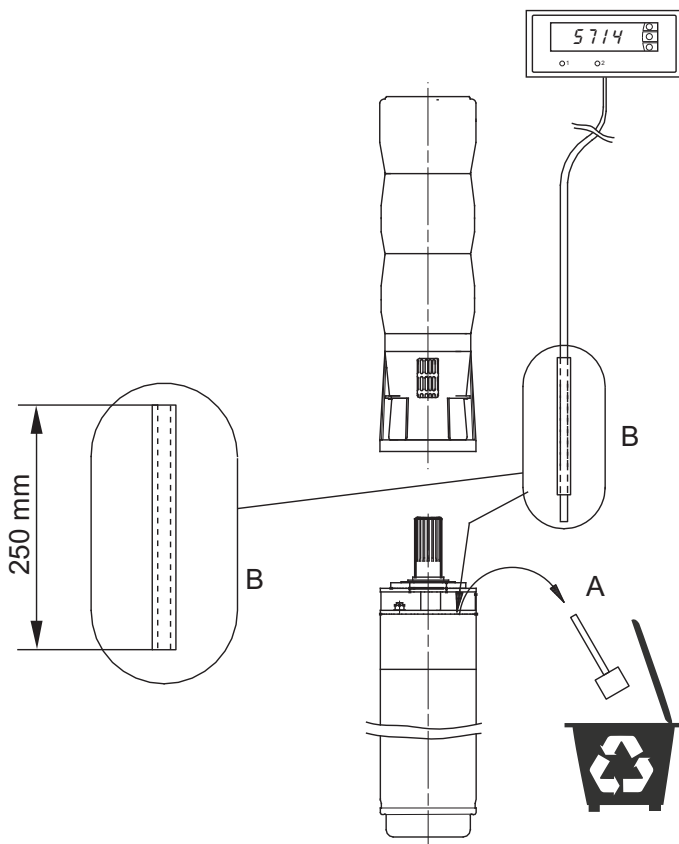


Fig. C MMS 6000 and MMS 8000

TM03 5454 3606

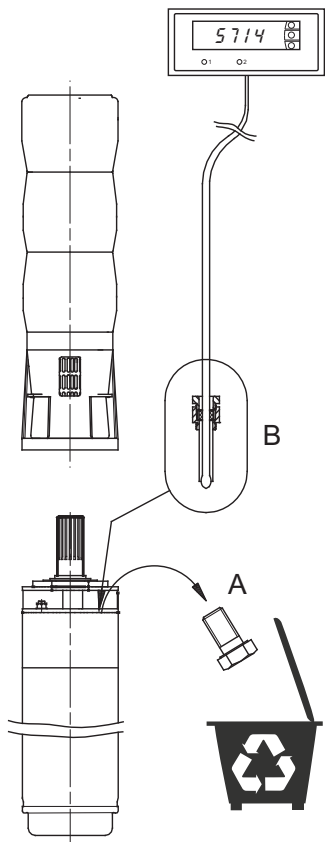


Fig. D MMS 10000

TM03 5450 3606

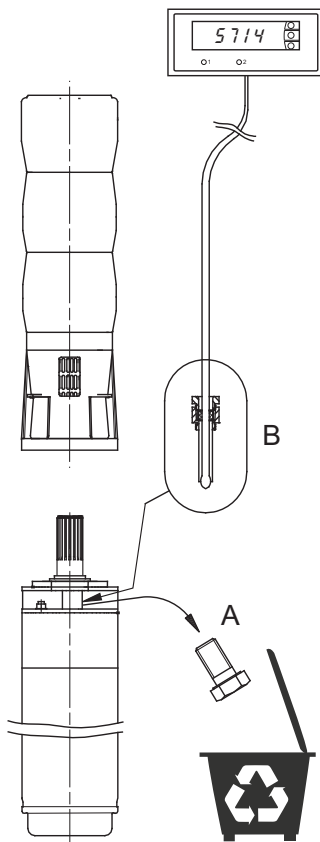


Fig. E MMS 12000

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